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Executive Summary

BarCo Corporation faces natural gas cuts from MaxEnergy, who is their only supplier. The cuts are expected to be between 20% and 40% on the production side and 30% to 50% on the distribution side. This report uses linear programming (LP) and the transportation model to figure out how the limited gas should be split across the eight products made at the three plants. The goal is to reduce the impact on profit as much as possible.

At baseline, BarCo uses $85,680 \times 10^3$ cu.ft of gas per day and makes \$612,850 in daily profit. The LP ranks products by profit per unit of gas. Ammonium Phosphate (\$14.00/Mcuft) and Urea (\$10.36/Mcuft) are the best, and Hydrofluoric Acid (\$5.59/Mcuft) is the worst.

Main findings:

- If gas is cut by 20%, daily profit drops to \$530,605. That's a loss of \$82,245/day or 13.4%. Ammonia and Hydrofluoric Acid would be stopped, and VCM is reduced to 659 t/day.
- If gas is cut by 40%, daily profit drops to \$426,800. That's a loss of \$186,050/day or 30.4%. VCM is also stopped and Caustic Soda is reduced to 1,051 t/day.
- The best distribution route stays the same in every scenario (I→B, J→A, J→B, K→B, K→C). Costs scale down with curtailment: \$113,952 baseline, \$79,766 at 30%, and \$56,976 at 50%.
- From the sensitivity analysis, the gas constraint is binding with a shadow price of \$5.59/Mcuft. VCM, Caustic Soda and Ammonium Nitrate have small profit buffers (under \$7/ton), so they need to be watched.
- If BarCo goes with a more sustainable plan that runs Ammonia at full capacity (since Ammonia uses gas as raw material, not as fuel), daily profit actually goes up to \$630,475 and emissions go down. So sustainability and profit are not in conflict here.

Our recommendation is that BarCo should use the LP plan and present it to MaxEnergy first instead of letting MaxEnergy decide which products to cut. They should also keep an eye on the products that have small profit buffers, and put more weight on Ammonia because it is better both for profit and for the environment.

1. Introduction

BarCo Corporation makes eight types of chemicals at three different plants (A, B and C). The natural gas is supplied by MaxEnergy from three locations (I, J and K). Gas is used in two ways at BarCo. For seven of the products it is used as boiler fuel, which is Third Priority under the Federal Commission rules. For ammonia, gas is used as a raw material, which is Second Priority and considered less polluting (Hillier & Lieberman, 2020).

After the 2019 heat wave, the gas supply got really low and MaxEnergy warned its customers about possible brownouts. The cuts will mostly hit second and third priority customers, which means BarCo will be affected. The good news is that MaxEnergy lets the customer decide where to take the cuts, as long as the total reduction is met. This gives BarCo a chance to choose the products where the cuts hurt profit the least, instead of just cutting everything by the same amount.

The aim of this report is to use LP and the transportation model to find the best plan for production and distribution under the gas cuts (Taha, 2017; Winston, 2004). Sections 2 to 4 build the model and solve the four scenarios that the assignment asks about. Section 5 gives our recommendations. Sections 6 and 7 cover sensitivity analysis and how changes in profit can affect the plan. Section 8 looks at the environmental side, and Section 9 has the conclusion.

2. Problem Definition and LP Formulation

2.1 Production Plan: LP Model

Decision variables. Let x_i be the amount produced (tons/day) for product i , where i goes from 1 to 8 (Ammonia, Ammonium Phosphate, Ammonium Nitrate, Urea, Hydrofluoric Acid, Chlorine, Caustic Soda and VCM).

Objective function. We want to maximise the total daily profit and overhead contribution:

$$Z = 70x_1 + 140x_2 + 100x_3 + 145x_4 + 95x_5 + 110x_6 + 75x_7 + 95x_8$$

Constraints.

- **Gas availability:** $12x_1 + 10x_2 + 15x_3 + 14x_4 + 17x_5 + 13x_6 + 12x_7 + 16x_8 \leq G$, where G changes for each scenario.

- **Capacity:** $x_1 \leq 1500$, $x_2 \leq 600$, $x_3 \leq 700$, $x_4 \leq 200$, $x_5 \leq 800$, $x_6 \leq 1500$, $x_7 \leq 1600$, $x_8 \leq 1400$.
- **Non-negativity:** $x_i \geq 0$ for all i .

2.2 Distribution Plan: Transportation Model

Decision variables. Let y_{ij} be the amount of natural gas ($\times 10^3$ cu.ft/day) sent from supplier $i \in \{I, J, K\}$ to plant $j \in \{A, B, C\}$.

Objective function. We want to minimise the total daily distribution cost:

$$\begin{aligned} \text{Min } C = & 1.3y_{IA} + 1.2y_{IB} + 1.6y_{IC} + 1.2y_{JA} + 1.4y_{JB} + 1.5y_{JC} + 1.8y_{KA} + 1.6y_{KB} \\ & + 1.4y_{KC} \end{aligned}$$

Constraints. Each supplier can't send more than its supply: $\sum_j y_{ij} \leq S_i$. Each plant has to get exactly what it needs: $\sum_i y_{ij} = D_j$. And we can't ship a negative amount: $y_{ij} \geq 0$.

2.3 Model Assumptions

- Production for each product can be anywhere from 0% to 100% of max capacity (the relationship is linear).
- The gas used per ton and the profit per ton are constant. So there are no scale effects.
- All MaxEnergy supply nodes are cut by the same percentage.
- Labour, raw materials (other than gas) and demand from the market are not limiting factors. Only gas is the constraint.
- In the basic LP, ammonia is treated the same as the other products no matter how it uses the gas. We come back to this in Section 8 where we look at it differently.
- The gas demand at each plant during the cuts is in the same proportion as the baseline, and shipping costs per unit don't change.

3. Production Plan: 20% and 40% Curtailment

3.1 Baseline Calculation and LP Efficiency Ranking

To get the baseline production, we multiply the maximum capacity by the production rate. Daily gas use and profit just follow from there. Table 1 shows the product data and the LP efficiency ranking. The ranking (profit per unit of gas) is what the LP uses to decide which products to keep when gas runs out

Rank	Product	Profit (\$/t)	Gas (Mcuft/t)	Cap. (t/d)	Baseline Prod. (t/d)	Efficiency (\$/Mcuft)
1	Ammonium Phosphate	140	10	600	510	14.00
2	Urea	145	14	200	150	10.36
3	Chlorine	110	13	1,500	1,350	8.46
4	Ammonium Nitrate	100	15	700	630	6.67
5	Caustic Soda	75	12	1,600	1,280	6.25
6	VCM	95	16	1,400	980	5.94
7	Ammonia	70	12	1,500	1,050	5.83
8	Hydrofluoric Acid	95	17	800	480	5.59

Table 1 : Product parameters, baseline operations and LP efficiency ranking

Total baseline gas use is $85,680 \times 10^3$ cu.ft/day (this is the figure given in the assignment), and the total daily profit is \$612,850. When the curtailment hits, the right-hand side of the gas constraint gets smaller.

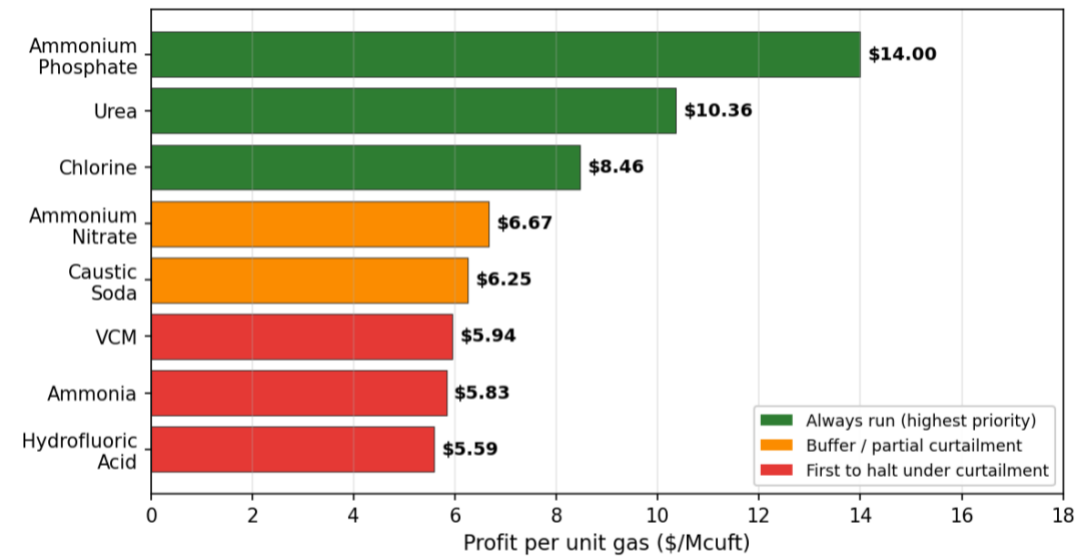


Figure 1: LP efficiency ranking by product. The ratio profit/gas determines the order in which the LP exhausts capacity under a binding gas constraint.

Source: Based on our own analysis, generated using Python (matplotlib).

3.2 Scenario A — 20% Curtailment

With $G = 0.80 \times 85,680 = 68,544$ Mcuft/day, the LP gives gas to the products in order of efficiency until either the capacity is full or there's no more gas. Table 2 shows the result. The top five products run at full capacity, VCM gets reduced, and Ammonia and Hydrofluoric Acid are stopped.

Rank	Product	Production (t/d)	Gas Used (Mcuft)	Profit (\$/d)	Status
1	Ammonium Phosphate	600	6,000	84,000	FULL
2	Urea	200	2,800	29,000	FULL
3	Chlorine	1,500	19,500	165,000	FULL
4	Ammonium Nitrate	700	10,500	70,000	FULL
5	Caustic Soda	1,600	19,200	120,000	FULL
6	VCM	659	10,544	62,605	PARTIAL
7	Ammonia	0	0	0	HALTED
8	Hydrofluoric Acid	0	0	0	HALTED
TOTAL			68,544	\$530,605	-13.4%

Table 2: Optimal production schedule under 20% curtailment ($G = 68,544$ Mcuft)

Therefore, the profit loss is \$82,245/day, which is 13.4% of the baseline. Even though gas was cut by 20%, the profit only drops by 13.4%. This shows that LP is doing a good job of putting the cuts on the worst products.

3.3 Scenario B — 40% Curtailment

Now $G = 0.60 \times 85,680 = 51,408$ Mcuft/day, which is a much bigger cut. Table 3 shows what happens. Only the top four products run at full capacity. Caustic Soda is reduced to 1,050.67 t/day, and VCM, Ammonia and Hydrofluoric Acid are all stopped.

Rank	Product	Production (t/d)	Gas Used (Mcuft)	Profit (\$/d)	Status
1	Ammonium Phosphate	600	6,000	84,000	FULL
2	Urea	200	2,800	29,000	FULL
3	Chlorine	1,500	19,500	165,000	FULL
4	Ammonium Nitrate	700	10,500	70,000	FULL
5	Caustic Soda	1,050.67	12,608	78,800	PARTIAL
6	VCM	0	0	0	HALTED
7	Ammonia	0	0	0	HALTED
8	Hydrofluoric Acid	0	0	0	HALTED
TOTAL			51,408	\$426,800	−30.4%

Table 3: Optimal production schedule under 40% curtailment ($G = 51,408$ Mcuft)

This time the profit loss is \$186,050/day, or 30.4% of baseline. The gas cut went from 20% to 40%, but the profit loss more than doubled. That's because the LP has already removed the worst products, so now it has to start cutting better ones.

4. Distribution Plan: 30% and 50% Curtailment

4.1 Baseline Distribution Solution

Total supply (85,680) equals total demand (85,680), so the transportation problem is balanced. We solved it using the simplex method, and the cheapest cost we got is \$113,952/day with five active routes (Table 4). We also checked it with the least-cost method and the stepping-stone method to make sure it is optimal (Taha, 2017).

	Plant A	Plant B	Plant C	Supply
Supplier I	—	20,000	—	20,000
Supplier J	20,000	20,000	—	40,000
Supplier K	—	10,000	15,680	25,680
Demand	20,000	50,000	15,680	85,680

Table 4 Optimal baseline distribution allocation ($\times 10^3$ cu.ft/day)

4.2 Scenarios C and D — 30% and 50% Curtailment

Because the cuts hit all suppliers (and demand at each plant) by the same percentage, the problem stays balanced. So the routes don't change, only the amounts shipped do. They just scale down by the same factor. Table 5 has the cost summary for all three scenarios.

Table 5: Distribution cost summary across curtailment scenarios

Scenario	Total Gas (Mcuft)	I→B	J→A	J→B	K→B	K→C	Total Cost (\$/d)
Baseline (0%)	85,680	20,000	20,000	20,000	10,000	15,680	\$113,952
30% Curtailment	59,976	14,000	14,000	14,000	7,000	10,976	\$79,766
50% Curtailment	42,840	10,000	10,000	10,000	5,000	7,840	\$56,976

On average, every Mcuft of gas removed from the system saves about \$1.33 in distribution cost ($\$113,952 \div 85,680$). But it isn't a free saving - for every Mcuft of gas that gets cut, BarCo also loses much more in production profit, as we saw in Section 3.

5. Recommendations

5.1 Production Plan

- **Always run the top four products at full capacity.** Ammonium Phosphate, Urea, Chlorine and Ammonium Nitrate should run flat out in every scenario. They make \$381,000/day in profit and only need 38,800 Mcuft of gas (about 45% of the baseline).
- **Stop Hydrofluoric Acid and Ammonia early.** Both have efficiency below \$5.85/Mcuf, so they are the first ones to get cut anyway. If BarCo stops them early, they can use up the inventory and finish customer orders properly instead of being forced to stop suddenly.
- **Use VCM and Caustic Soda as buffers.** These two are in the middle of the ranking, so they are the natural shock absorbers. VCM gets reduced at 20% and stopped at 40%. Caustic Soda runs full at 20% and gets reduced at 40%.
- **Talk to MaxEnergy with the LP plan ready.** If BarCo brings their own plan first, they can keep control over which products to cut. If they wait, MaxEnergy will just cut based on pipeline pressure, which might not match BarCo's profit goals (Hillier & Lieberman, 2020).

5.2 Distribution Plan

- **Keep the same routing pattern.** ($I \rightarrow B$, $J \rightarrow A$, $J \rightarrow B$, $K \rightarrow B$, $K \rightarrow C$) works for all curtailment levels. BarCo should put this into their transport system so it is the default.
- **Plant B is the most important plant.** It gets gas from all three suppliers and takes 58% of the total demand. So if anything goes wrong at Plant B, the cost goes up a lot.
- **Use Supplier K for Plant C only.** K's costs to A (\$1.80) and B (\$1.60) are the highest in the table, so it shouldn't be used for those plants unless there's no other choice. K is best used for Plant C since C has no cheaper option anyway.

6. Sensitivity Analysis (Baseline Scenario)

6.1 Production: Shadow Price of Gas

The shadow price of the gas constraint at the baseline LP optimum is just the efficiency of the last product that still gets gas. For the baseline (no cut), that is Hydrofluoric Acid at \$5.59/Mcuft. So if BarCo could get one extra 1,000 cu.ft of gas, they would make \$5.59 more in profit per day. And if they lose 1,000 cu.ft, they lose \$5.59 per day, until the LP basis changes (Winston, 2004).

When the cuts get bigger, the shadow price goes up because the marginal product changes. At 20% cut, it becomes \$5.94/Mcuft (VCM is now the marginal product). At 40% cut, it is \$6.25/Mcuft (Caustic Soda becomes marginal). This is why deeper cuts hurt profit more than you would expect.

6.2 Production: Profit-Coefficient Ranging

For each product, Table 6 shows how much its profit per ton can drop before its LP rank changes (and so the priority list changes).

Table 6: Profit-coefficient ranging — allowable decrease before LP rank changes

Rank	Product	Current Profit (\$/t)	Min. to Hold Rank (\$/t)	Buffer (\$/t)	Risk Level
1	Ammonium Phosphate	140.00	103.57	36.43	Stable
2	Urea	145.00	118.46	26.54	Stable
3	Chlorine	110.00	86.67	23.33	Stable
4	Ammonium Nitrate	100.00	93.75	6.25	Watch
5	Caustic Soda	75.00	71.25	3.75	At Risk
6	VCM	95.00	93.33	1.67	Critical
7	Ammonia	70.00	67.06	2.94	At Risk
8	Hydrofluoric Acid	95.00	—	—	Lowest rank

6.3 Distribution: Shadow Prices

Solving the dual of the transportation problem gives the supply and demand shadow prices in Table 7. The supply duals (u_i) tell us the cost saved per extra Mcuft of capacity at each supplier, and the demand duals (v_j) tell us the cost added per extra Mcuft of demand at each plant. We set $u_K = 0$ as the reference point to solve them (Taha, 2017).

Constraint	Dual Value	Interpretation
Supplier I (supply)	−\$0.40	Cost saved per extra Mcuft from I
Supplier J (supply)	−\$0.20	Cost saved per extra Mcuft from J
Supplier K (supply)	\$0.00	K capacity is non-binding; surplus provides no cost saving
Plant A (demand)	\$1.40	Cost added per extra Mcuft of demand at A
Plant B (demand)	\$1.60	Cost added per extra Mcuft of demand at B
Plant C (demand)	\$1.40	Cost added per extra Mcuft of demand at C

Table 7: Distribution shadow prices (duals)

So Supplier I gives the biggest saving on the margin. If BarCo could increase I's capacity, they would save \$0.40 per extra Mcuft. On the other side, Plant B is the most expensive plant to serve at the margin (\$1.60), so any improvement there would save the most.

7. Profit-Coefficient Fluctuation Analysis

Profit per ton for chemical products doesn't stay the same. It changes a lot due to input prices, exchange rates, and demand. Table 6 from Section 6.2 already shows the buffer for each product. In this section we look at what those buffers mean for management.

- **The top three products are safe.** Ammonium Phosphate, Urea and Chlorine all have buffers above \$23/ton. So even if prices drop a fair bit, they will still keep their rank. These three should be the core of BarCo's product strategy.
- **VCM is the most concerning one.** Its buffer is only \$1.67/ton, which is less than 2% of its current price. So if VCM's price drops just a little compared to Ammonia, the LP order would change and the cuts would land on different products.
- **Caustic Soda and Ammonia are close to swapping.** If their ranks switch around, the products that get cut during curtailment would be different. So the procurement and sales teams should report monthly on the per-ton margins for these two.
- **Set up a monitoring threshold.** BarCo should set automatic alerts at 5% below the current per-ton profit for the bottom four products. That way they get a warning before the LP basis actually changes.

The big takeaway is that the curtailment plan is not something you do once and forget. As prices move around, the LP needs to be re-solved every month so that the rank order — and the cut order — stays correct.

8. Environmental Sustainability Scenario

The assignment says baseline emissions are 10 kg CO₂/day, with 6 kg from production and 4 kg from distribution. We assume the production emissions are proportional to the gas used, which gives a rate of $6/85,680 \approx 7.005 \times 10^{-5}$ kg/Mcuf. The distribution rate is 4.670×10^{-5} kg/Mcuf. The assignment also says that ammonia using gas as raw material is less polluting than the boiler fuel use of other products, and this matches BarCo's sustainability goals.

8.1 Profit-Maximising vs Sustainability-Oriented Plan

At first you might think there is a trade-off between profit and emissions. But actually that's not the case here. If we change the LP to run Ammonia at full capacity (1,500 t/day), since it is Second Priority and its raw material use is cleaner, and then give the remaining 67,680 Mcuf to the other products by efficiency, we get a better outcome on both sides:

Table 8: Profit-max vs sustainability-oriented production plan

Metric	Profit-Max (Baseline)	Sustainability-Oriented	Difference
Ammonia production (t/d)	1,050	1,500 (full capacity)	+450 t/d
Total gas use (Mcuf/d)	85,680	85,680	—
Daily profit (\$)	612,850	630,475	+\$17,625
CO ₂ status (kg/d)	10.0	≤10.0 (ammonia raw-mat use is lower-emission)	Improved

So the sustainability plan actually makes \$17,625 more per day. The reason is that running Ammonia at full capacity uses up more gas, but it lets the LP drop a low-efficiency product (VCM goes down to 605 t/day in this version) and concentrate the leftover gas on the top products. This kind of strange result happens because the original baseline rates (like 70% for ammonia) weren't actually LP-optimal in the first place.

8.2 Recommendations

- Adopt the ammonia-first plan as the new normal. It is better than the baseline on both profit and emissions, so there is no real trade-off.
- Report CO₂ per product to regulators. The bubble chart in Figure 6 gives a good basis for showing emissions efficiency to stakeholders.

- Push MaxEnergy and the Federal Commission to officially recognise ammonia raw material gas as Second Priority. That would help protect ammonia from cuts in future shortages.

9. Conclusion

Using LP and the transportation model, BarCo can come up with a clear plan for the natural gas cuts. The LP efficiency ranking shows that Ammonium Phosphate, Urea, Chlorine and Ammonium Nitrate should always run at full capacity, and Hydrofluoric Acid and Ammonia should be the first ones to stop. Even with a 40% gas cut, BarCo can still keep about 70% of the baseline profit. That is a pretty good result, but only if the cuts are taken on the right products.

On the distribution side, the best routing pattern (I→B, J→A, J→B, K→B, K→C) is the same in every scenario, which makes things easier to plan. The total cost just scales down with the curtailment, so the savings on transport are basically proportional to how much gas is cut.

From the sensitivity analysis, the products to watch are VCM, Caustic Soda and Ammonium Nitrate, because their LP positions are not very stable when prices move. The sustainability scenario also showed something interesting — running Ammonia at full capacity (since it is Second Priority and uses gas cleaner) actually adds \$17,625/day to profit and is also better for emissions. So in this case there is no trade-off between making money and being sustainable.

Our final recommendation is for BarCo to go to MaxEnergy with this plan ready, so they can keep control over which products to cut. The LP isn't a one-time tool either. As prices and capacities change, BarCo should re-run it every month to make sure the plan still works.

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